

Haodong Liang

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Education

University of California, Davis, USA. Ph.D. Electrical and Computer Engineering	Sep 2023 – Present
• Advisors: Lifeng Lai, Krishnakumar Balasubramanian • GPA: 4.0/4.0	
University of Michigan, Ann Arbor, USA. MSc. Statistics	Sep 2021 – May 2023
• GPA: 3.95/4.0	
Xiamen University, China. BSc. Mathematical Statistics	Sep 2016 – Jun 2020
• GPA: 3.64/4.0. Major GPA: 3.8/4.0.	

Research Interests

My primary research focuses on developing a theoretical and mechanistic understanding of how Transformer models learn and execute algorithms in context. I am also interested in reinforcement learning, continual learning, and differential privacy.

Research Projects

Provable In-context Learning Capabilities of Transformer Models	Jan 2024 - Present UC Davis
• Studied the theoretical and mechanistic foundations of how Transformers learn and execute algorithms in context, supported by empirical validations. • Established that Transformers can handle endogeneity in the in-context linear regression setting. • Established that Transformers can implement policy-improvement algorithms in the in-context reinforcement learning setting.	
Differential Privacy under Endogeneity	Mar 2025 - Sep 2025 UC Davis
• Proposed a noisy two-stage gradient descent algorithm that addresses endogeneity while guaranteeing ρ-zero-concentrated differential privacy. • Established finite-sample convergence rate for the proposed method. • Demonstrated practical privacy-utility trade-offs on both synthetic and real-world datasets.	

Selected Publications

Transformers Provably Implement In-Context Reinforcement Learning with Policy Improvement	2026
<i>H. Liang</i> , L. Lai arXiv preprint (2026) Paper	
Differentially Private Two-Stage Gradient Descent for Instrumental Variable Regression	2025
<i>H. Liang</i> , Y. Jin, K. Balasubramanian, L. Lai The Fourteenth International Conference on Learning Representations (ICLR 2026) Paper	
Transformers Handle Endogeneity in In-Context Linear Regression	2024
<i>H. Liang</i> , K. Balasubramanian, L. Lai The Thirteenth International Conference on Learning Representations (ICLR 2025) Paper	
ASGARD is A Single-cell Guided Pipeline to Aid Repurposing of Drugs	2023
B. He, Y. Xiao, <i>H. Liang</i> , Q. Huang, Y. Du, Y. Li, D. Garmire, D. Sun, L. Garmire Nature Communications 14, 993 (2023) Paper	

Professional Services

Reviewer: ICLR (2026), NeurIPS (2026), IEEE Transactions on Information Theory.

Teaching Assistant: EEC 161 (Applied Probability), EEC 018 (Digital System Design), EEC 266 (Information Theory and Coding).

Consultant: CSCAR (Consulting for Statistics, Computing & Analytics Research).

Mentor: Maxwell Lin (Princeton University), Kaleb Kokott (UC Davis).

Honors and Awards

Shu Lin Graduate Student Conference Travel Award	UC Davis, 2026
Summer Graduate Student Researcher Award	UC Davis, 2025
University Graduate Program Fellowship	UC Davis, 2023

Work Experience

BaishanCloud Technology	Jul 2020 - Jul 2021
Data Analyst - Full time	Xiamen, Fujian, China

- Designed optimization models and operational metrics for network bandwidth allocation, minimizing infrastructure cost while satisfying service quality constraints.
- Developed Python-based data pipelines integrated with internal APIs to automatically collect, process, and monitor network utilization metrics on a daily basis.
- Achieved 14% cost reduction (~\$1.2M annual savings).

Skills

Languages: English, Mandarin.

Programmings: Python, R, Matlab, C++ , Julia.

Hobbies: Running, Badminton, Hiking.